

Optimization of Graph Squares - Proofs

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1 Formulations

Let known constants $g_{ij} \in \{0, 1\}$ come from the adjacency matrix of the square A_G and let $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. The variables in the program are x_{ij} , y_{ikj} , z_{ij} where x_{ij} form the ij -th entry of the adjacency matrix of the root A_H .

1.1 Integer Program

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (1)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (3)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (4)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (5)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (7)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (8)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (9)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (10)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (12)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (13)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (14)$$

1.2 Linear Program

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (15)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (16)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (17)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (18)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (19)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (20)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (21)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (22)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (23)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (24)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (25)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (26)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (27)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (28)$$

1.3 Max Linear Program

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (29)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (30)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (31)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (32)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (33)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (34)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (35)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (36)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (37)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (38)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (39)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (40)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (41)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (42)$$

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (43)$$

2 Two-Edge-Triangle Proof

Let x be an optimal solution to the linear relaxation defined by Equations (29)–(43). If $g_{ij} = 0$, then for every $k \in \{1, \dots, n\}$,

$$x_{ik} + x_{kj} = 1.$$

2.1 Integer Program

Lemma 1. *Consider the triangle with three vertices i, j and k where edges ik and kj are present and ij is absent in square G . Then for root G , the integer program defined by Equations (1)–(14) must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. As ij is absent in G , $g_{ij} = 0$ such that Equation (6) gives $z_{ij} \leq 0$ and Equation (7) gives $z_{ij} \geq 0$ such that $z_{ij} = 0$. Therefore Equation (5) gives

$$0 = \sum_{a=1}^n y_{iaj} + x_{ij} = \sum_{a=1}^n x_{ia} x_{aj} + x_{ij},$$

where the second equality follows from the product constraints Equations (8)–(10) together with the integrality of $x \in \{0, 1\}$ by Equation (1) and Equation (11). Since all variables are non-negative, 0 only holds when $x_{ij} = 0$ and all of $x_{ia}x_{aj} = 0$ and using the integrality of x , this means that at most one of x_{ia} , x_{aj} is equal to 1 and the other must be 0, which can be equivalently written as $x_{ia} + x_{aj} \leq 1$. Taking $a = k$ gives $x_{ik} + x_{kj} \leq 1$ as required. $\square$

2.2 Linear Program

Lemma 2. *Consider the triangle with three vertices i , j and k where edges ik and kj are present and ij is absent in square G . Then for root G , the linear program defined by Equations (15)–(28) must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. Since $g_{ij} = 0$, Equations (20)–(21) imply that $z_{ij} = 0$. Equation (19) therefore gives

$$0 = z_{ij} = \sum_{a=1}^n y_{iaj} + x_{ij}.$$

Since all variables are nonnegative by Equations (15)–(17), every term in the above sum must be equal to zero with $x_{ij} = 0$ and $y_{iaj} = 0$ for all a . Thus applying Equation (24) to $y_{ikj} = 0$ gives

$$0 = y_{ikj} \geq x_{ik} + x_{kj} - 1,$$

and hence $x_{ik} + x_{kj} \leq 1$ as required. $\square$

2.3 Max Linear Program

Proposition 1. *Consider the maximum linear relaxation defined by Equations (29)–(43) on the graph induced by the three vertices $\{i, j, k\}$. Suppose*

$$g_{ik} = 1, \quad g_{kj} = 1, \quad g_{ij} = 0,$$

then there always exists an optimal, feasible solution satisfying $x_{ik} + x_{kj} = 1$.

Proof. Since $g_{ij} = 0$, Equations (34)–(35) thus imply that $z_{ij} = 0$. Equation (33) therefore gives

$$0 = \sum_{t \in \{i, j, k\}} y_{itj} + x_{ij} = y_{iij} + y_{ijj} + y_{ikj} + x_{ij} = y_{ikj} + x_{ij}, \quad (44)$$

since Equations (36)–(37) together with Equation (42) give that $y_{iij} \leq x_{ii} = 0$ and $y_{ijj} \leq x_{jj} = 0$ and as all variables are non-negative by Equations (29)–(31), it follows that $y_{iij} = 0$, $y_{ijj} = 0$. Therefore Equation (44) gives

$$y_{ikj} = 0, \quad x_{ij} = 0.$$

By contradiction, assume that x_{ik} and x_{kj} give a optimal, feasible solution to the maximal linear program where $x_{ik} + x_{kj} = 1 - \delta$ and $0 \leq \delta \leq \varepsilon$ such that $x_{ik} + x_{kj} < 1$. Consider the perturbation $x'_{ik} = x_{ik} + \epsilon$ such that $x'_{ik} + x_{kj} = 1$. $\square$

Proposition 2. *Consider the maximum linear relaxation defined by Equations (29)–(43) on the square graph G . Suppose*

$$g_{ik} = 1, \quad g_{kj} = 1, \quad g_{ij} = 0,$$

and let (x, y, z) be a feasible solution satisfying $x_{ik} + x_{kj} < 1$. Then in order for there to exist another feasible solution (x', y', z') such that $\sum_{u, v \in \{i, j, k\}} x'_{uv} > \sum_{u, v \in \{i, j, k\}} x_{uv}$ G must be infinitely large.

Proof. Using Proposition 1, any three vertices that satisfy the maximum linear relaxation defined by Equations (29)–(43) will always satisfy $x_{ik} + x_{kj} = 1$. Hence the only way $x_{ik} + x_{kj} < 1$ is if $\square$

Lemma 3. *Let x be a feasible solution to the maximum linear relaxation defined by Equations (29)–(43). Suppose there exists vertices i, j and k such that*

$$g_{ik} = 1, \quad g_{kj} = 1, \quad g_{ij} = 0 \quad \text{and} \quad x_{ik} + x_{kj} < 1.$$

Then there exists another feasible solution (x', y', z') satisfying $\sum_{u,v} x'_{uv} > \sum_{u,v} x_{uv}$.

Proof. $\square$

Theorem 1. *Let x be an optimal solution to the linear relaxation defined by Equations (29)–(43). If $g_{ij} = 0$, then for every $k \in \{1, \dots, n\}$,*

$$x_{ik} + x_{kj} = 1.$$

Proof. Using Lemma 2 $x_{ik} + x_{kj} \leq 1$ and using Proposition 3 $x_{ik} + x_{kj} \geq 1$ such that $x_{ik} + x_{kj} = 1$ as required. $\square$